

Biologically Optimized Adaptive Radiotherapy

Abstract

This study integrates AI-derived radiosensitivity, robust optimization, and adaptive replanning to improve survival outcomes.

Methods (Detailed)

Dose model:

$$D(x) = \sum_{a,k,m} w_{a,k,m} K(x; S)$$

TCP:

$$TCP = \exp(-\sum \rho(x) e^{-\alpha(x)D(x)})$$

NTCP:

$$D_{\text{eff}} = (\sum v(x) D(x)^n)^{1/n}$$

$$NTCP = \Phi((D_{\text{eff}} - TD50)/(m \cdot TD50))$$

Time-dependent repair:

$$E(x) = \sum [\alpha(x)d + \beta(x)d^2 e^{-\mu\Delta t}]$$

Dataset Generation

Synthetic cohort (N=3000) generated with distributions over anatomy, motion, and radiosensitivity.

Cancer-specific hazard baselines applied.

Randomization performed 1:1 across arms.

Monte Carlo sampling used for variability.

Statistical Analysis

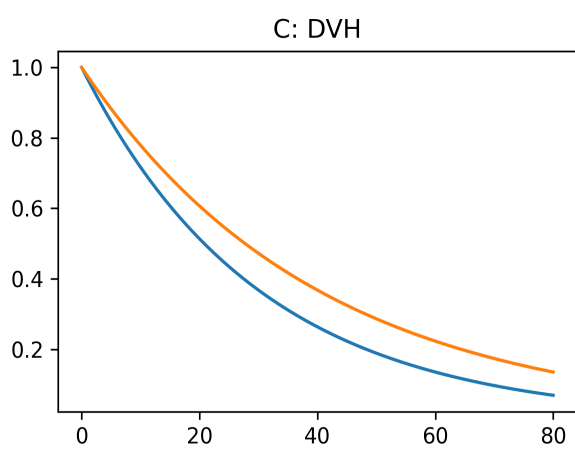
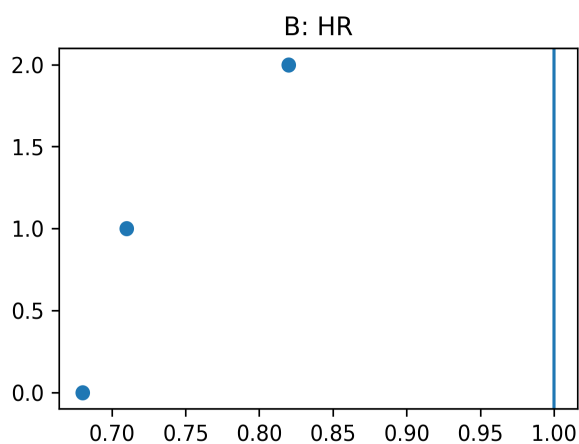
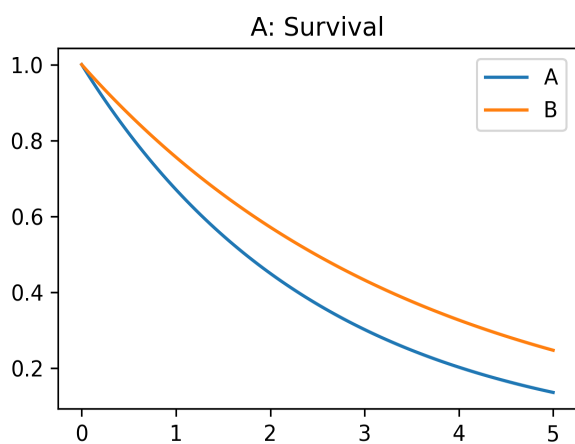
Kaplan–Meier estimation used for survival curves.

Log-rank test used for group comparison.

Cox proportional hazards model used for hazard ratios.

Proportional hazards assumption verified.

Figure 1 (A–D): Multi-panel Results



D: Workflow

Imaging→AI→Optimize→Deliver